ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Final Project Report

SWE 4304: Software Project Lab I

Project Title

IUT CafeCache

Team Members

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1. Problem Statement

IUT-CafeCache addresses several critical operational and financial challenges faced by institutional cafeteria management systems:

i. Inefficient Payment Processing: Traditional cafeteria systems rely on manual cash transactions or basic card systems, leading to long queues, transaction delays, and increased operational overhead during peak hours.

ii. Lack of User Authentication and Security: Without robust authentication mechanisms, cafeteria systems are vulnerable to unauthorized access, fraudulent transactions, and potential financial losses. There is a critical need for secure user verification and access control.

iii. Absence of Digital Token/Recharge Management: Current systems lack an integrated digital token system, making it difficult for users to manage prepaid balance, track recharges, and maintain transparent transaction histories.

iv. Limited Analytics and Business Insights: Cafeteria operators lack comprehensive sales analytics and performance metrics, making it challenging to optimize inventory, forecast demand, and make data-driven business decisions.

v. Inadequate Feedback Mechanism: There is no structured system to collect, manage, and analyze user feedback, preventing the cafeteria from identifying service improvement opportunities and addressing customer concerns systematically.

By integrating comprehensive authentication, digital token management, sales analytics, and feedback systems, IUT-CafeCache aims to create a streamlined, secure, and data-driven cafeteria management solution tailored for institutional environments.

2. Potential Solution

IUT-CafeCache proposes an integrated digital cafeteria management platform that streamlines payment processing, enhances security, and provides operational insights through automation.

The solution combines:

i. A secure authentication framework to control user access and prevent unauthorized transactions

ii. A digital token-based payment system for seamless balance management and recharges

iii. Analytics capabilities to track sales and optimize business decisions

iv. Feedback mechanisms for continuous service improvement

By consolidating these functionalities into a single, cohesive system, IUT-CafeCache eliminates manual processes, enhances transaction security, and empowers administrators with actionable business intelligence—all tailored for institutional cafeteria environments.

3. Implemented Features

1) Authentication Manager

Problem Solved: Securing access to the cafeteria system and preventing unauthorized transactions.

Details:

i. Implements user login with username and password validation

ii. Verifies user credentials against stored user database

iii. Manages user sessions post-authentication

iv. Controls access to system functions based on user authentication status

v. Prevents unauthenticated users from accessing transaction features

2) User Management

Problem Solved: Maintaining user profiles and tracking individual account information.

Details:

i. Stores user information including username, password, and role

ii. Maintains user balance/account details

iii. Reads and writes user data to persistent storage (users.txt)

iv. Supports user profile creation and retrieval

v. Tracks individual user account status

3) Token System

Problem Solved: Creating a digital payment mechanism for cafeteria transactions.

Details:

i. Generates unique tokens for transactions

ii. Associates tokens with user information and transaction amounts

iii. Stores token data persistently (tokens.txt)

iv. Enables token-based payment processing

v. Maintains token records for transaction tracking

4) Recharge Management

Problem Solved: Managing user balance top-ups and prepaid account management.

Details:

i. Allows users to submit recharge requests with desired amounts

ii. Stores recharge requests in a queue (recharge_requests.txt)

iii. Provides admin interface to view pending recharge requests

iv. Implements approval workflow for recharge validation

v. Updates user balance upon recharge approval

5) Sales Analytics

Problem Solved: Tracking transactions and providing business insights to administrators.

Details:

i. Collects transaction data

ii. Generates sales reports for administrators

iii. Analyzes transaction patterns and revenue metrics

iv. Provides summary statistics on cafeteria operations

v. Enables data-driven decision making for inventory and pricing

6) Feedback System

Problem Solved: Collecting and managing user feedback for service improvement.

Details:

Allows users to submit feedback/complaints about cafeteria services

Stores feedback persistently (feedback.txt)

Maintains feedback records with user information and timestamps

Provides admin interface to review submitted feedback

Enables categorization and analysis of feedback for service improvements

4. Tools and Technology

The development of the IUT CafeCache system utilises a set of tools. These tools assist our team to be well equipped to build the proposed software.

- **Programming Language (C++)**: Chosen for its performance in handling complex data processing and fast execution time.
- **Development Environment (VS Code)**: This provides a versatile environment for coding, debugging, and efficient management of the C++ project files.
- **Version Control (Git)**: Used for local version control, allowing the team to track changes and manage the history of the codebase.
- **Version Control Hosting (GitHub)**: Serves as the remote repository for collaboration, code backup, and managing the project's evolution across the team.
- **Documentation (Google Docs)**: It enables collaborative report writing, project documentation, and maintaining records of requirements and architectural decisions.

5. Application Domain

IUT-CafeCache is designed specifically for the institutional cafeteria management sector, with primary applicability in the following domains:

i. Educational Institutions: Universities, colleges, and technical institutes that operate centralized cafeteria facilities serving large student and staff populations.

ii. Institutional Food Service Operations: Cafeterias within organizations that require systematic payment processing, user authentication, and transaction management for food and beverage services.

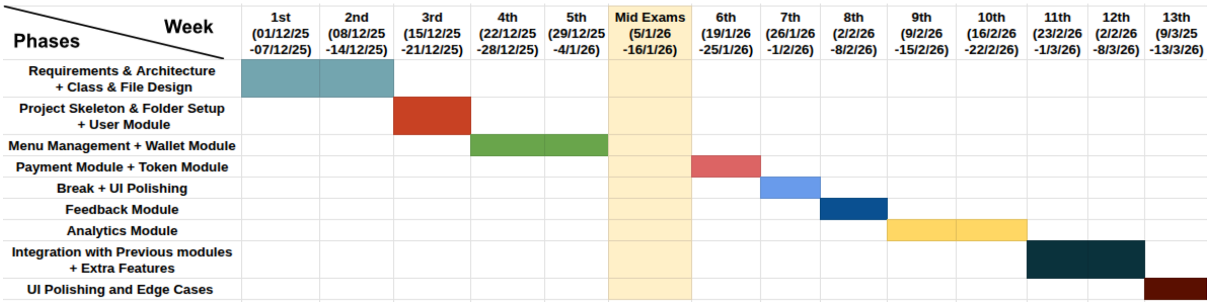
iii. Prepaid Meal Systems: Institutions implementing digital wallet or prepaid balance systems for streamlined transaction processing and reduced cash handling.

iv. Digital Payment and Analytics: Organizations seeking to transition from traditional cash-based cafeteria operations to digitized payment systems with comprehensive transaction analytics and reporting capabilities.

v. Customer Feedback Management: Institutional cafeterias requiring structured mechanisms to collect, analyze, and act upon customer feedback for continuous service quality improvement.

6. Project Timeline

The project timeline is organized over 13 weeks and is structured to address dependencies, starting with architectural foundation and concluding with final analytics integration and UI polishing.



Phase	**Duration**	**Activities and Deliverables**
Requirements & Architecture	1st and 2nd week	Gather and analyze system requirements. Define project scope and design overall architecture including initial class and file structure.
Project Skeleton & Setup+User Module	3rd week	Set up project skeleton, repository structure, and implement the basic user module (authentication and user management).

Menu Management Module	4th and 5th week	Develop menu management system and implement wallet module for handling user balance and transactions.
Payment and Token Module	6th week	Implement payment processing and token generation modules for order handling.
Break+UI Polishing	7th week	Short break and perform initial UI polishing and usability improvements.
Feedback Module	8th week	Develop feedback module allowing users to submit reviews and ratings.
Analytics Module	9th and 10th Week	Implement analytics module to track system usage, transactions, and generate basic insights.
Integration with previous modules+Extra features	11th and 12th week	Integrate all previously developed modules and add extra features. Perform integration testing.
UI Polishing and Edge Cases	13th Week	Final UI polishing, handle edge cases, bug fixing, and prepare final project delivery.

7. Contribution

Name	Worked in modules
Mahmudul Hossain Samir(230042101)	User & Authentication Module, File Manager Module, Feedback Module, UI Polishing(Partial), Menu Management Module(Partial)
Raiyan Al Arafat(230042134)	Payment & Token Module, Sales Analytics Module(Partial), UI Polishing(Partial)
Tamjidur Rahman(230042109)	Recharge Management(Wallet) Module, Sales Analytics Module(Partial), Menu Management Module(Partial)

8. Future Work

IUT-CafeCache can be enhanced and scaled with the following plans:

i. Mobile Application: Develop iOS and Android apps for balance checking, recharge requests, and transaction history access.

ii. Advanced Payment Integration: Support multiple payment gateways including UPI, credit cards, digital wallets, and QR code payments.

iii. Predictive Analytics: Implement machine learning for demand forecasting, inventory optimization, and fraud detection.

iv Real-Time Dashboard: Create an interactive web-based admin dashboard for transaction monitoring and advanced reporting.

v. Automated Notifications: Deploy email/SMS alerts for balance warnings, recharge status, and promotional updates.

vi. Enhanced Security: Implement two-factor authentication, end-to-end encryption, and advanced audit logging.

vii. Loyalty Program: Introduce points-based rewards, tiered memberships, and referral bonuses.

viii. Cloud Migration: Transition to cloud infrastructure for scalability, backup, and multi-institution support.

9. Source Repository

Github Link